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Thermally driven countercurrent plasma centrifuge

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ABSTRACT

Renewed interest in plasma isotope separation stems from the need to produce certain isotopes more inexpensively. These include isotopes of groups I and II, rare earth and some other elements of the periodic table. While gas centrifuge cascades, developed for uranium enrichment, are capable of a wider range of separation applications, they are limited to elements that can form volatile gaseous compounds at room temperature. In the absence of a suitable compound, more expensive separation techniques are used, consuming considerable electrical power for relatively low throughputs. We believe that a new type of plasma centrifuge can be a viable alternative.

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Introduction

Today, more than 100 stable isotopes of 28 chemical elements can be obtained by means of the gas centrifuge separation method.^[1,2] However, alkali metals, alkaline earth elements, rare earths, and light inert gases are not suitable candidates for isotope separation by the GC method. While the isotopic products of such elements are widely used in various industrial, medical, agricultural, and scientific research,^a their production relies on far more expensive separation technologies, including electromagnetic separation, which is costly to operate because of its relatively large electrical power consumption and limited overall material throughput.

Calcium isotopes are of particular interest in several fields. Isotopes ⁴²Ca, ⁴⁴Ca, ⁴⁶Ca and ⁴⁸Ca are widely used in clinical research and nutritional studies for the measurement of calcium absorption in human bones. The neutron-abundant isotope, ⁴⁸Ca is used in nuclear physics to create super heavy elements.^[3] Another potential application of calcium isotopes is related to the determination of the nature of the neutrino mass and its absolute magnitude, which is a priority challenge for modern elementary particle physics. Researchers^[4] have proposed the use of cryogenic scintillation detectors on a substrate of ⁴⁰Ca¹⁰⁰MoO₄ crystals, in which the calcium compound is depleted in the ⁴⁸Ca isotope. Researchers in this field expect that it should allow them to achieve the sensitivity necessary for observation of the neutrino-less double beta-decay.

Potassium isotopes are also used in a wide range of fields: ³⁹K and ⁴¹K, are used in agrochemical studies; ³⁹K is applied in development of magnetometers for the biomedical research; ⁴¹K is needed in radiopharmaceutical production to obtain the ⁴²K radioisotope by the ⁴¹K(n, γ)⁴²K reaction, where ⁴²K is used to determine the total content of potassium and the volume of circulating red blood cells in the body.^[5]

Currently, isotopes of potassium and calcium are obtained by the universal but extremely expensive method of electromagnetic separation.^[6] However, the low throughput of this method and the high cost of produced isotopes motivate the development of an alternative, more cost-effective enrichment method.

In the early 1970s, the use of a fast-rotating plasma was proposed for the separation of isotopes.^[7,8] In spite of the results obtained for the radial separation effect, which were comparable to those achieved by other contemporary separation methods, its practical use encountered difficulties with the continuous extraction of the separated fractions.

Increased effective separation can be achieved by combining the radial separation effect with an axial counterflow of the heavy and light fractions along the separation vessel that contains the gaseous medium. More detailed theoretical and physical aspects, describing the multiplication of the radial separation effect along the length of column-type device, can be found in.^[1] Note that circulation not only increases the degree of separation but also facilitates the

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^aThere is an increasing need globally for certain types of isotopes to support science research, such as neutrinoless double beta decay experiments, solar neutrino detection and studies about dark matter in the universe

separation process by establishing product and waste zones at the ends of the column.

In the mid-1960s, Bonnevier proposed the multiplication of the radial separation effect in a PC with crossed constant electric and axial magnetic fields.^[7] The first calculation of the multiplication effect along the length of the column in such a system was published in.^[9]

While there are several different methods that can be employed to induce plasma rotation,^[7,8,10] the most appropriate for the PC described here is a system utilizing an external rotating magnetic field.^[11] The mechanism that induces radial motion resembles the action of an induction motor. This approach, where effective plasma rotation is induced using a high frequency (HF) rotating electromagnetic field, has been shown experimentally in.^[12,13] The measured velocity in^[12] correlated well with those calculated in the framework of the magnetohydrodynamic (MHD) approximation.

Electromagnetic methods have been proposed to create the required counter-current flow, or circulation, by the action of a traveling magnetic field (TMF).^[14,15] Through the creation of notional PC models, calculation of the associated operating parameters showed that high axial separation ratios can be achieved in such systems.^[11,15] In these preliminary models, the axial countercurrent flow of the medium was created by an external delay line, where the inductive coil created an axial TMF along the axis of the

separation chamber.^[15] While the arrangement is calculated to produce a magnetic field sufficient in strength to induce the axial countercurrent flow in the contained, rotating medium, the approach also significantly complicates that PC model's design and increases the calculated material and electrical operating costs.

In this paper, we propose an alternative method to establish the countercurrent flow by a simpler and cheaper thermal method, where the lower end of the PC is cooled. This approach is similar to that employed in some types of GC.^[16] We also evaluate the circulation speeds that can be excited at real values of the axial temperature gradients.

Rotational motion of a medium in a straight circular cylinder, which is closed at both ends, creates two symmetric circulation flows of four "vortices" (see Fig. 1a).^[9]

A similar situation also occurs in a PC with weak ionization of the contained medium. At both end boundary layers, the plasma will move to the axis under the action of a radial pressure gradient that is not balanced by the centrifugal force. The closure of the flow in such a system takes place through the main volume of the medium. In principle, this type of centrifuge can work as a separation device, while one of the fractions will be concentrated in the central part of the working chamber and the other is concentrated at both end zones (Fig. 1b). However, as a practical separation

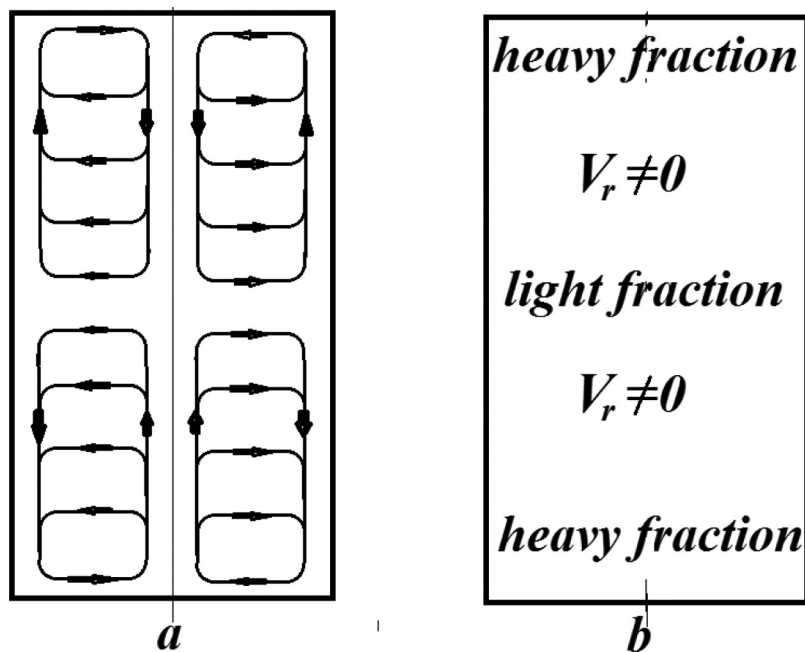


Figure 1. Stream lines and distribution of heavy and light fractions in a closed cylinder with two braking ends for a symmetric axial temperature distribution: (a) the shape of the circulation flows; (b) a qualitative picture of the redistribution of component concentrations along the PC height.

device, this type of system complicates either the supply of feed flow or the removal of the separated fractions. The feed flow would have to be introduced at two zones in the PC's separation chamber, while withdrawal of the heavy fraction would have to be carried out at both ends of the device. Furthermore, this configuration corresponds to a radial flow $V_r \neq 0$ directed toward the periphery in the main flow region (see Fig. 1a). This leads to a slowdown of the azimuthal flow by inertial forces, preventing the achievement of high rotation speeds and therefore high separation factors.

Circulation flow with two "vortices" and a radial velocity in the working chamber close to zero are implemented in modern GCs by means of a braking element at one end^[1] (see Fig. 2a).

This particular circulation pattern is due to the presence of a scoop in the upper region, which acts as a break on the circulating medium. In Fig. 2, the scoop is modeled as a flat stationary disk. The scoop causes a radial flow to the central axis, which leads to the closure of the axial flow circuit through the boundary layer at the lower end where the centrifugal force exceeds the radial pressure gradient. In this case, there is a relatively weak braking action on the rotating azimuthal flow ($V_\phi \approx \omega r$, where ω is the angular velocity of the rotor and r is the radial coordinate), which results in establishing the axial countercurrent flow, necessary for the multiplication of the radial separation effect along the column length. To maintain a continuous separation

process, the heavy and light fractions at both ends of the column require extraction.

Overview of design

Taking the PC design described in,^[15] HF fields are used to both partially ionize and rotate the medium. However, in the study presented here, we have removed the TMF system used to induce axial circulation of the partially ionized medium in previous studies.

A detailed description of the PC's design, with a rotating magnetic field, as well as the required technical parameters of the HF-device is given in^[11] and^[15] Fig. 3 is a simplified diagram of the modeled PC. In a practical device, an arrangement of azimuthally located rod conductors can be used to create a rotating magnetic field, which is fed from a system of HF generators operating at the same frequency, with each phased-shifted with respect to the other. The simplified schematic in Fig. 3a shows only two current-carrying rods. The configuration produces a resultant rotating dipole magnetic field at an angular speed ω_0 in the cylindrical volume of the working chamber.

To calculate the HF electromagnetic field generated by the current-carrying rods, the actual distribution of surface currents is usually replaced by a continuous rotating current layer. This idealized distribution is shown in Fig. 3b, where the rotating magnetic field drags the plasma in the azimuthal direction, due to the

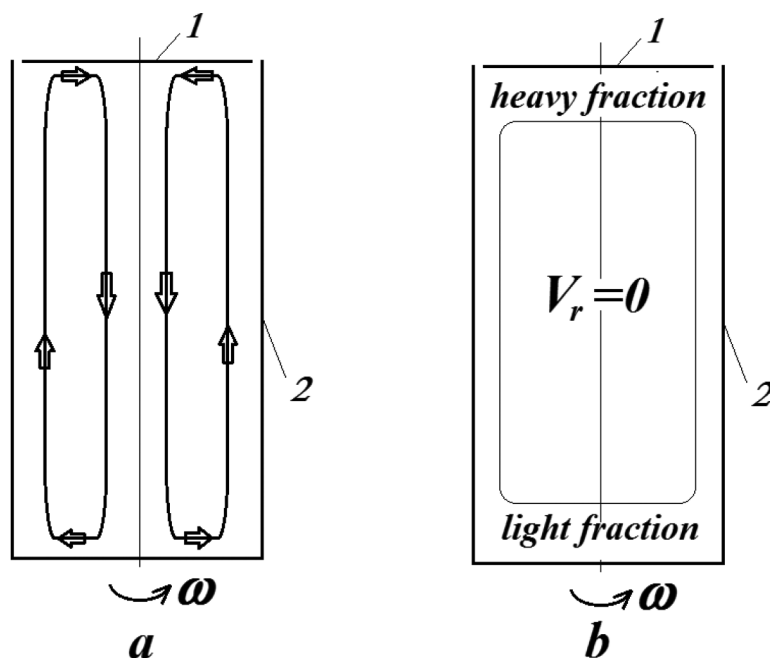


Figure 2. A GC with one braking element in the upper part of the column. Major components are as follows: 1 – fixed braking disk, or “scoop” and 2 – rotating cylinder. Figure 2(a) shows the streamlines; Fig. (b) demonstrates the qualitative distribution of fraction concentrations.

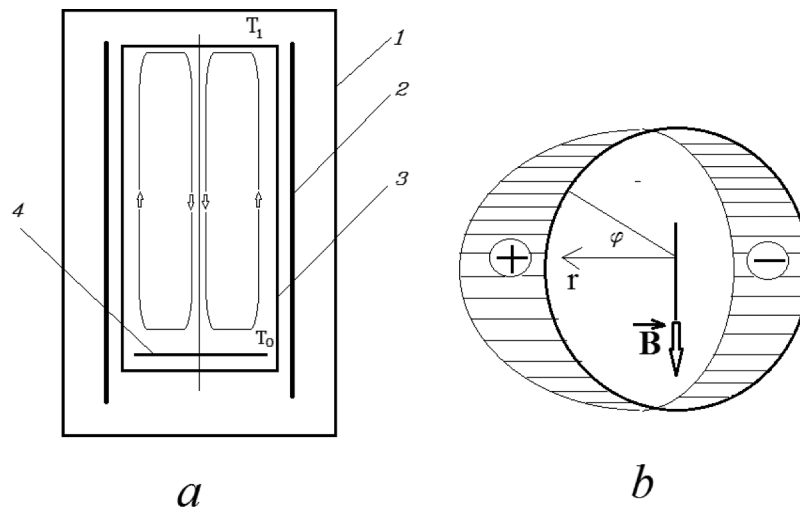


Figure 3. A simplified diagram of the device (*a*) and the instantaneous density distribution of surface currents that create a rotating magnetic field (*b*): 1 is an external vacuum chamber; 2 are the axial current-carrying rods (others are not visible in the figure); 3 is the working chamber, where separation occurs; 4 is the lower cooling disk (the associated cooling system is not shown).

phased interaction of the inductive axial electric current and the radial magnetic field.

We do not consider the process of plasma generation in detail in this paper, but assume that the applied HF fields are sufficient to create and maintain partial ionization of the medium. In a practical device, it is likely that an additional HF source may be necessary. However, creation and maintenance of such a discharge is confirmed in experimental studies.^[12,13] In the framework of the MHD approximation, the calculated plasma drag on the centrifugal force excitation scheme, proposed in,^[15] demonstrated satisfactory agreement with those achieved experimentally in^[12] and.^[13]

As emphasized above, for multiplication of the radial separation effect along the length of a PC, the use of the closed ends of the cylinder as the braking elements is ineffective, due to the radial plasma flow in the main volume of the PC's separation chamber. In order to prevent the closure of the circulation flow through the central part of the chamber, it becomes necessary to change the conditions substantially at one of the ends. In the model PC presented in this paper, we have used the bottom end for this purpose. For this, the average volumetric centrifugal force in that end's boundary layer must exceed the force associated with the radial pressure gradient induced by rotation of plasma in the main volume. This effect can be achieved by increasing the mass density of the gaseous medium at the PC's lower end PC with the incorporation of a cooled disk located at the bottom of the working chamber (see item 4 in Fig. 3). This approach is a significant departure from earlier PC studies and is presented here for the first time.

Practical implementation of the proposed scheme can be performed by cooling the lower end of the working chamber to a temperature T_0 (see Fig. 3a), while maintaining a higher temperature T_1 on the upper end.

The new PC model uses these conditions to substantiate the premise that the applied temperature differential between the top and bottom of the separation chamber will change the direction of the flow in the boundary layer at the lower stationary end. Accurate numerical simulation of the gas dynamics of the radially bounded rotational flow in the thin boundary layer above the surface of a stationary disk, in the presence of an axial temperature gradient, while taking nonlinear hydrodynamic phenomena into account, is complex and time-consuming.

We have attempted to obtain analytical dependences for the hydrodynamic characteristics of the circulation flow in the framework of the quasi-linear approach. For this, only the non-linear centrifugal term has been taken into account in the equations of motion in the projection on the radial axis r , while only the linear solution of the motion equation in the projection on the azimuthal coordinate φ is considered. Such problems arise, for example, in the presence of gas suction on the surface of the disk.^[17]

Problem statement and governing equations

The mechanism for controlling the radial flow in the boundary layers on rotating disks, in the presence of the main rotating flow, has been described in detail elsewhere.^[18] For making estimates, we consider plasma rotation with a constant angular velocity ω_1 above the

disk surface in the presence of an axial temperature gradient. The distributions of the azimuthal and radial velocity components, as well as the temperature profiles in the boundary layer, are shown in Fig. 4 in the isothermal case of a hot (a) and a cooled (b) disk surfaces.

The qualitative distributions of the radial pressure gradient are shown in Fig. 5a,b.

Figures 4 and 5 illustrate the physical mechanism for changing the direction of the radial flow in the boundary layer. In 5a, where the temperature and density are independent of the axial coordinate, the radial pressure gradient exceeds the centrifugal force and a radial flow arises in the boundary layer directed to the axis. The corresponding direction of the radial flow velocity is illustrated by the diagram in Fig. 4a. The opposite inequality holds in 4b. In the case of increased cooling

of the disk (case b) due to an increase in the density of cold gas near the end surface, the centrifugal force prevails over the pressure gradient. This leads to a change in the direction of the radial flow, since the heavier gas is more strongly displaced toward the periphery, as shown in the radial velocity diagram, Fig. 4b. Thus, in the stationary regime, which is studied in this work, the azimuthal plasma velocity does not depend on the axial coordinate in the main volume and sharply decreases to zero on the end surfaces. In turn, the radial velocity near the upper end is directed toward the axis, and near the lower end – in the opposite direction, in the central zone it is zero.

For the analysis of plasma rotational motion, it is necessary to know the value of the azimuthal bulk electromagnetic force acting on the conductive medium,

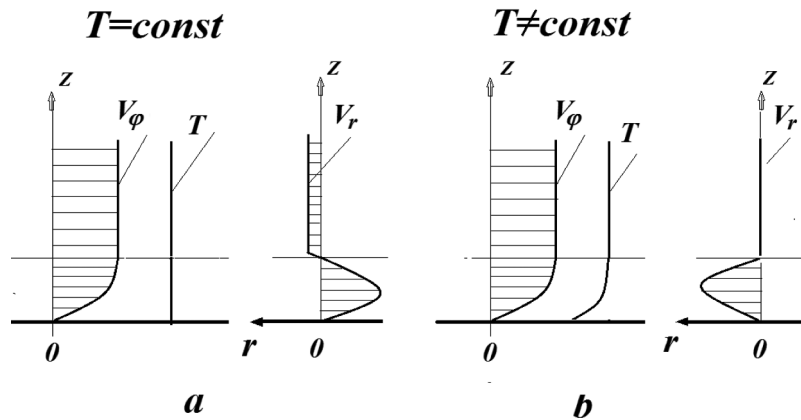


Figure 4. The radial and azimuthal velocity profiles in the boundary layer in the case of an extended disk: a) $dT/dZ = 0$ ($n = T_1/T_2 = 1$); b) $dT/dZ \neq 0$ ($n > 1$).

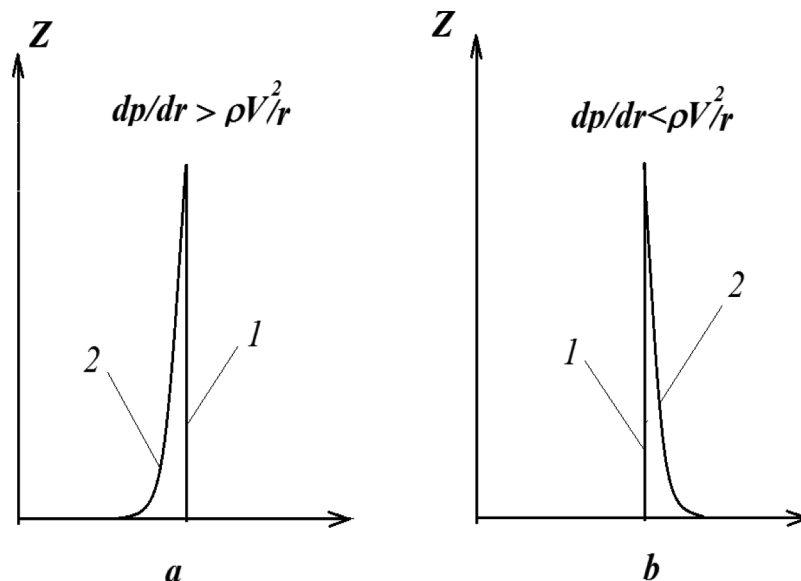


Figure 5. The axial pressure gradient distribution dp/dr (1) and the centrifugal force $\rho V_\phi^2/r$ (2) in the cases of (a) and (b) in Fig. 4.

due to the interaction between the axial electric current and the radial component of a magnetic field. For this purpose, we replace the discrete distribution of the longitudinal currents in the external conductors by a continuous rotating current layer, distributed over the surface of the cylinder,^[15] in the following equation:

$$I_z = I_{z0} \exp[i(\omega_0 t - \varphi)], \quad (1)$$

where I_{z0} is the amplitude value of the axial current surface density; φ is the azimuthal coordinate and ω_0 is the angular velocity of magnetic field rotation. A plot of the surface current density distribution is shown in Fig. 3b. The characteristics of the field in a vacuum are described by the following formulas:

$$B_r = B_0 \cos(\omega_0 t - \varphi), \quad B_\varphi = B_0 \sin(\omega_0 t - \varphi), \quad (2)$$

where B_r and B_φ are the radial and azimuthal components of a magnetic field, respectively, and B_0 is the peak amplitude value of the magnetic field. The frequencies to be used in our plasma centrifuge are closed to $\omega_0 \approx 10^5$ 1/s ($\nu \approx 10^4$ Hz).^[12,13] Note that these frequencies are much lower than that for collision frequencies of plasma particles ν_{ei} and ν_{en} . Following,^[15] we will not take into account the induced magnetic field. In this approximation, the amplitude of the field is independent of the radial coordinate. The associated axial electric current is described by the following equation:

$$j_z = \sigma \omega_0 r B_0 \cos(\omega_0 t - \varphi), \quad (3)$$

where σ is the plasma conductivity.

The equations describing stationary hydrodynamic and thermal boundary layers on an extended disk, in the presence of the radial pressure gradient caused by rotation of the main flow with a certain angular velocity ω_1 , take the following forms:

$$\rho \left(v_r \frac{\partial v_r}{\partial r} + v_z \frac{\partial v_r}{\partial z} - \frac{v_\varphi^2}{r} \right) = -\frac{1}{\rho} \frac{\partial p}{\partial r} + \frac{\partial}{\partial z} \left(\eta \frac{\partial v_r}{\partial z} \right) - \langle j_z B_\varphi \rangle \quad (4)$$

$$\rho \left(v_r \frac{\partial v_\varphi}{\partial r} + v_z \frac{\partial v_\varphi}{\partial z} - \frac{v_r v_\varphi}{r} \right) = \frac{\partial}{\partial z} \left(\eta \frac{\partial v_\varphi}{\partial z} \right) + \langle j_z B_r \rangle, \quad (5)$$

$$\frac{\partial}{\partial r} (\rho v_r) + \frac{\partial}{\partial z} (\rho v_z) = 0 \quad (6)$$

$$\rho c_p \left(v_r \frac{\partial T}{\partial r} + v_z \frac{\partial T}{\partial z} \right) = \frac{\partial}{\partial z} \kappa \frac{\partial T}{\partial z} \quad (7)$$

where the symbol $\langle \rangle$ signifies field period averaging; z is an axial coordinate in the direction away from the disk surface; v_r , v_φ and v_z are, respectively, the period

averaged HF fields in the radial, azimuthal and axial velocity components of the medium; κ is the thermal conductivity coefficient; $\eta = \rho \nu$ is dynamic viscosity; p is pressure; T is temperature and c_p is specific heat capacity at constant pressure. Note that axial electric currents are closed at the ends of the device, as is the case in a conventional induction motor. The second term in the right-hand side of Eq. (5), i.e. $\langle j_z B_r \rangle$ describes the average volumetric electromagnetic force over a period that accelerates plasma in the azimuthal direction. The system of Eqs. (4) – (7) does not contain the equation of motion in the direction of the z axis, since the latter serves only to determine the weak dependence of pressure on the axial coordinate in the boundary layer.^[18]

To the system of Eqs. (4)–(7), the equation of state can be added in the following form:

$$p = \rho \Re T / \mu, \quad (8)$$

where μ is a molar mass and $\Re$ is the universal gas constant.

The approximation of a small change in density along the radius can be considered when the following inequality holds:

$$\chi = \frac{\mu V_\varphi^{*2}}{2 \Re T} < 1 \quad (9)$$

where V_φ^* is the characteristic value of the azimuthal gas velocity.

Note that neglecting viscous dissipation with respect to convective heat transfer in the energy Eq. (7) is valid under the conditions: $\omega_1^2 R^{*2} / c_p T^* \ll 1$, where T^* is the characteristic temperature of a gas in the flow and R^* is the characteristic radius. An estimation of this parameter for a monatomic gas with molar mass $\mu = 20$ kg/kmol at a temperature of $T^* = 2 \cdot 10^3$ K can be made. Assuming the Poisson constant $\gamma = 5/3$; the angular velocity $\omega_1 = 10^4$ 1/s; $R^* = 0.05$ m and $c_p = 1000$ J/kg·K, we obtain the following approximate value for $\omega_1^2 R^{*2} / c_p T^* \approx 0.1$.

We assume that the gas temperature does not depend on the radial coordinate, which is in accordance with neglecting viscous heating in the energy Eq. (7). An assumption can also be made that the heat capacity of the gas will be independent of temperature. Note that the real separation column is limited in a radial direction. However, the thickness of the Ekman boundary layers is usually much smaller than the radial dimension of the separation device. Therefore, for estimation of the circulation flow, we consider the boundary layer at the extended end of the column. The system of Eqs. (4) – (7), in the presence of uniform gas suction on the surface

of the disk with speed, will be solved under the following boundary conditions:

$$z = 0 \quad v_r = 0, \quad v_\varphi = 0, \quad v_z = -k, \quad T = T_0, \quad (10)$$

$$z \rightarrow \delta \quad v_r \rightarrow 0, \quad v_\varphi \rightarrow \omega_1 r, \quad T \rightarrow T_1 \quad (11)$$

The Dorodnitsyn-Howard transformation is now introduced so that we may execute the transformation of the axial component of the velocity, as follows^[19,20]:

$$Z_0 = \int_0^z \left(\frac{\rho(z)}{\rho_1} \right) dz, \quad v_{z1} = v_z \rho / \rho_1 \quad (12), (13)$$

We also assume the following relationships^[18]:

$$v_r = rF(Z_0), \quad v_\varphi = rG(Z_0), \quad T = T_0 + (T_1 - T_0)t(Z_0) \quad (14)$$

The coefficients of dynamic viscosity and thermal conductivity change proportionally to the first power of temperature, i.e. $\eta = \eta_1 T/T_1$ and $\kappa = \kappa_1 (T/T_1)$.^[21] Here η_1 and κ_1 are the coefficients of dynamic viscosity and thermal conductivity in the external flow, respectively.

Recalling that the external flow rotates at a constant angular velocity ω_1 and has density ρ_1 , the following condition is fulfilled in the boundary layers^[11]:

$$\frac{dp}{dr} = \rho_1 \omega_1^2 r \quad (15)$$

The force acting on the gas volume element in the boundary layer associated with the pressure gradient is directed toward the axis and depends on the density in the external flow ρ_1 and the angular velocity ω_1 . The centrifugal force is directed to the periphery and is determined by the axial density $\rho(z)$ and azimuthal velocity distributions $v_\varphi(z)$. It becomes possible to change the intensity and circulation directions by varying the thermophysical parameters of the model. By cooling the disk surface, thereby increasing the gas density in the boundary layer, it is possible to increase the average centrifugal force over the layer cross-section and overcome the effect of the pressure gradient. Using the calculation method developed in,^[18] which is based on the reduction of partial differential equations of the boundary layer to a system of ordinary differential equations, the following system of equations are obtained under the condition $S = \sigma B_0^2 / \rho_1 \omega_1 < 1$:

$$F^2 + v_{z1} F' - G^2 = -\omega_1^2 \left[\frac{1}{n} + \left(1 - \frac{1}{n}\right)t \right] + v_1 F'', \quad (16)$$

$$2FG + v_{z1} G' = v_1 G'', \quad (17)$$

$$2F + v_{z1}' = 0, \quad (18)$$

$$v_{z1} \theta' = \chi_1 t'', \quad (19)$$

where $\chi_1 = \kappa_1 / \rho_1 c_p$ is the thermal diffusivity; $v_1 = \eta_1 / \rho_1$ is the kinematic viscosity in the external flow and $n = T_1 / T_0$. The prime denotes the **derivative** with respect to the variable, Z_0 .

To solve the system of Eqs. (16) – (19) we substitute the function $v_{z0} = v_{z1} + k_1$, where $k_1 = \rho_0 k / \rho_1$ and ρ_0 is the density of the conducting gas near the disk's surface.

Considering the case where there is a large suction value (i.e. $v_{z1} \ll k_1$) and taking Eq. (18) and introducing the dimensionless functions: $g = G/\omega_1$, $f = F/\omega_1$ and the variable $Z = Z_0 \sqrt{\omega_1 / v_1}$, and $K_1 = k_1 / \sqrt{v_1 \omega_1}$, we derive the following set of equations:

$$-g^2 + m^2 \left[\frac{1}{n} + \left(1 - \frac{1}{n}\right)t \right] = f'' + K_1 f', \quad (20)$$

$$g'' + K_1 g' = 0, \quad (21)$$

$$t'' + K_1 Pr t' = 0, \quad (22)$$

where $Pr = v_1 / \chi_1$ is the Prandtl number and the prime indicates differentiation by Z .

Integrating the system of Eqs. (20) – (22), we find the following expressions:

$$t(Z) = 1 - \exp(-K_1 Pr Z), \quad (23)$$

$$g(Z) = 1 - \exp(-K_1 Z) \quad (24)$$

The function $f(Z)$ is determined by solving the following differential equation:

$$f'' + K_1 f' - Nf = -[1 - \exp(-K_1 Z)]^2 + \frac{1}{n} [n + (1 - n) \exp(-K_1 Pr Z)] \quad (25)$$

The solution for $f(Z)$ is as follows:

$$f(Z) = \exp(-K_1 Z) \left\{ \frac{(n-1)}{(Pr(Pr-1)K_1^2 - N)n} [1 - \exp(K_0 Z - Pr K_1 Z)] + \frac{1}{2K_1^2} [1 - \exp(-K_1 Z)] - \frac{2Z}{K_1} \right\} \quad (26)$$

Major results

Figure 6,7 and 8 are the calculated results for the radial velocity component in the boundary layer for the suction parameter $K_1 = 1$, various values of the parameter $n = T_1 / T_0$ and the Prandtl number Pr .

The results demonstrate that for the relatively large Prandtl numbers (e.g. $Pr = 0.7$ in Fig. 6), with an increase in the parameter n , there is a weak tendency to slow the gas flow directed to the axis of a PC. This phenomenon can be explained as follows: for an increase in the n parameter, more intensive cooling of the medium occurs adjacent to the disk. As a consequence, there is an increase in centrifugal force. This counteracts the force associated with the pressure gradient and the radial flow toward the device's axis decreases. At the same time, with a decrease in the Prandtl number, this tendency intensifies and at $Pr = 0.3$ (see Fig. 7) there is a change in the flow direction with an increase in the degree of cooling of the disk surface. This means that for values of the parameter Pr there exists a value ($n^* \approx 2$) at which the radial pressure gradient is balanced by centrifugal force.

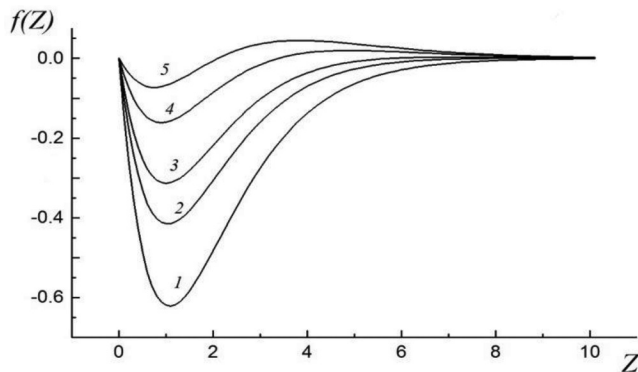


Figure 6. Dependences of the radial velocity component in the boundary layer at the Prandtl number $Pr = 0.7$ and various values of the n parameter. 1 – 1; 2–1.5; 3–2; 4–4; 5–10.

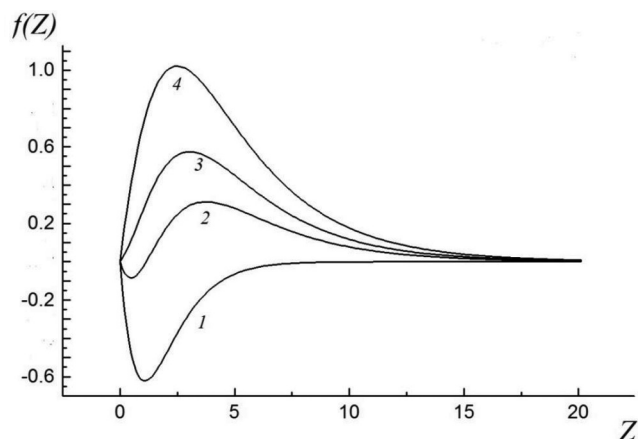


Figure 7. Dependences of the radial velocity component in the boundary layer at the Prandtl number $Pr = 0.3$, for values of the n parameter, where for curve 1 $n = 1$; 2 $n = 1.5$; 3 $n = 2$ and 4 $n = 4$.

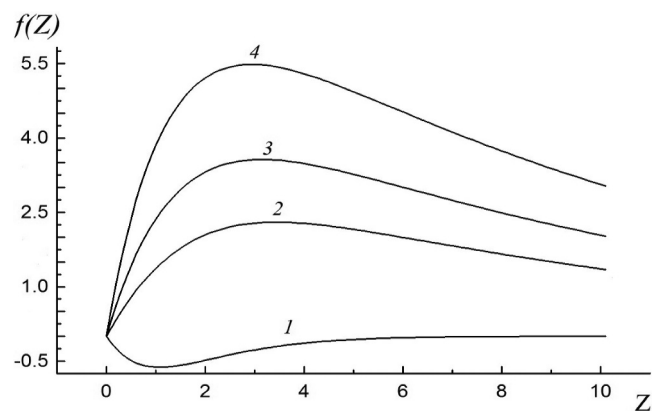


Figure 8. Dependences of the radial velocity component in the boundary layer at the Prandtl number $Pr = 0.1$ and for the same values of the n parameter as in Fig. 7.

With the Prandtl number $Pr = 0.1$, the absolute value of the radial flow velocity, directed to the periphery, increases by an order of a magnitude compared to the same value in the absence of the disk surface cooling.

These dependences are explained by the axial temperature distribution characteristics in the boundary layer, which are described by Eq. (23). This relationship shows that as the Prandtl number Pr decreases, the thickness of the thermal boundary layer increases, leading to more efficient cooling of the gas near the disk's surface and strengthening the mechanism of centrifugal rejection of the medium toward the periphery.

As described in,^[1,22] the characteristic velocity of the axial circulation flow, corresponding to the maximum enrichment factor in a separation column, can be estimated as $V_z^* \approx \rho D / \langle \rho \rangle R^*$, where D is the self-diffusion coefficient of the separated isotopes; $\langle \rho \rangle$ is the characteristic average density of a working substance and R^* is the radius of a working chamber.

For example, using this approach, we can estimate the value of V_z^* for the separation of neon isotopes. Assuming $\rho D \approx \eta \approx 10^{-4} \text{ kg/m} \cdot \text{s}$ (where η is the dynamic viscosity coefficient, $\langle \rho \rangle = 10^{-4} \text{ kg/m}^3$ and $R^* = 0.05 \text{ m}$), we obtain an axial flow of $V_z^* \approx 20 \text{ m/s}$. We can also estimate the value of the axial circulation velocity in accordance with the results obtained for $Pr = 0.3$ and for $n = 4$ (see Fig. 6). Considering that $V_z \approx V_r \delta / R^*$, where the thickness of the hydrodynamic boundary layer $\delta \approx \sqrt{\eta / \langle \rho \rangle \omega_1}$, for $\omega_1 = 10^4 \text{ 1/s}$ we find that $V_z \approx 10^2 \text{ m/s}$. Note that transverse magnetic fields $B_0 \approx 10^{-2} \text{ T}$ are required to achieve angular velocities $\omega_1 \approx 10^4 \text{ 1/s}$.^[15]

These estimates demonstrate that the circulation, excited by cooling the lower end of a working chamber, is sufficient for the effective multiplication of the radial separation factor and that the circulation can be controlled by changing the axial temperature gradient.

For this reason, it can be argued that in the case of separation of isotopes that do not have convenient gaseous compounds at room temperature, the plasma centrifuge can provide more attractive prices than the electromagnetic method of separation.^[6]

In conclusion, we estimate the value of the parameter S . Assuming the plasma conductivity is equal to $\sigma = 10^3 \text{ 1/}\Omega\text{-m}$ and $B_0 = 10^{-3} T$, we obtain the S parameter value $S \approx 10^{-3}$.

Conclusions

There is a growing demand for enriched stable isotopes for a wide range of applications. It is well known that the cost of stable isotopes that cannot be produced by means of a conventional GC is very high. The PC described in this paper potentially offers an improved means of meeting the demand for certain types isotopes that require far more costly separation technologies. However, many important basic plasma phenomena in such devices remain unexplored. In this paper, we have proposed and modeled a PC, which for the first time, can establish the required circulation flows for effective separation by means of an axial temperature gradient. We have also shown that decreasing the associated Prandtl numbers in the PC, by deep cooling of the boundary layer, is the most effective means for exciting countercurrent flow, thereby multiplying the enrichment effect.

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